AS

CONTACT

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GITHUB
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LOCATION
DSMax Skyscape, Bangalore 560037

SKILLS

Languages
Java 8-21 · JavaScript · TypeScript · SQL · Shell

Backend
Spring Boot · Spring Security · Microservices · REST APIs · Hibernate / JPA · J2EE · Node.js · Express

Streaming & Messaging
Apache Kafka · Apache Flink · Kafka Streams · Hadoop · RabbitMQ · MQTT / Mosquitto

Data
Azure SQL · Cosmos DB · Redis · Elasticsearch · OpenSearch · MongoDB · NoSQL

DevOps
Docker · Kubernetes · Istio / mTLS · Jenkins · CI/CD · Maven · Gradle · Git

Security & Quality
HMAC / Webhook Auth · API Gateway Policies · Code Signing · SonarQube · Fortify · Black Duck · TDD

AI & Integration
MCP Servers · AI Agents · Webhooks

AMRITA **SINGH**

Staff Software Engineer

Backend & Distributed Systems · Event Streaming · Engineering Leadership

100M+	14+	10	7M
NOTIFICATIONS DELIVERED DAILY	YEARS BUILDING SOFTWARE	ENGINEERS LED	SUBSCRIPTIONS MIGRATED

PROFILE

Staff Software Engineer with 14+ years of experience designing, building and scaling backend platforms. At Walmart Global Tech, I lead engineering on Walmart Marketplace's notifications platform, which delivers 100M+ notifications a day on Java, Spring Boot, Kafka and Apache Flink. I'm the platform's design authority and subject-matter expert, I've led teams of up to 10 engineers, and I've delivered large data migrations, security and compliance rollouts, and AI integrations. I lead RCAs for executive escalations and align architects, IAM and product teams around practical designs.

EXPERIENCE

Present
Feb 2022

Staff Software Engineer

Walmart Global Tech · Bangalore

PROJECT
Notifications Platform
Java, Spring Boot, Flink, Kafka, Elasticsearch, Cosmos DB, Hadoop

PROJECT
Subscription Management Service
Java, Spring Boot, Azure SQL, Redis, Kafka

PROJECT
Customer-Seller Communications Platform
REST API, Java, Spring Boot, Microservices, Elasticsearch, Cosmos DB

ROLE & RESPONSIBILITIES

- Design authority and primary reviewer for platform-level architecture changes across the notifications platform's services.
- Subject-matter expert for the platform; represent it in platform-stability calls and executive escalations.
- Lead root-cause analysis for high-visibility incidents and present findings and remediation plans to engineering leadership.
- Drive alignment with architects, Principal Engineers, IAM, Product and partner teams on cross-team designs.
- Identify delivery risks early, manage external dependencies and communicate trade-offs to leadership.
- Mentor engineers, onboard and ramp up team members, and run a regular review and unblocker cadence.

- Lead engineering on Walmart Marketplace's event-driven, multi-channel notifications platform, which delivers 100M+ notifications a day to sellers, suppliers and partners with real-time streaming on Apache Flink, Kafka and Hadoop.
- Led a team of 10 engineers to design and deliver platform features end to end, from requirements and design reviews through production rollout.

Frontend

Angular 2–8 | AngularJS | Electron | HTML5 | CSS / SCSS | Bootstrap

STRENGTHS

- System design at scale
- Technical leadership & mentoring
- Large-scale data migrations
- Incident RCA & escalations
- Security-first engineering
- Cross-team collaboration

- Worked on the ETL for the Subscription Management Service (Azure SQL and Redis) that migrated about 7 million subscriptions, live by quarter-end as targeted, and supported re-migration to close gaps found in the sync service.
- Drove Workload Identity (mTLS) adoption across 20+ Flink deployments and multiple Kafka connections: wrote the phased deployment plan for 30 workloads and added the shared-library support that let downstream services adopt it without custom code.
- Designed the principal-ID approach for email notifications, presented the constraints and trade-offs to architects and aligned on a practical interim solution with IAM and Principal Engineers.
- Led the RCA for an executive escalation on new-order delivery dates, tracing failures across Kafka offsets, Flink job state and downstream delivery, and drove follow-on hardening.
- Proposed a redesign of 18 independent Flink jobs into 3 tiered applications (bloom-filter pre-rejection, async I/O with circuit breakers), projecting 3–5× throughput and a 60–70% cut in operational overhead, plus a Kafka Streams option for low-latency workloads.
- Built and onboarded the platform's first MCP server, letting sellers manage webhooks through AI assistants in natural language, plus an AI agent for webhook notification workflows. The demo led to a related item in product intake.
- Moved email delivery onto the enterprise templated email platform within the target timeline by managing an external team's dependency risk, which cut third-party email (Mailgun) costs.
- Delivered a service's migration from the legacy hosting platform to Kubernetes with Istio service mesh as part of the decommission program, and diagnosed an Istio mTLS misconfiguration that was causing Kafka timeouts in staging.
- Built a live notification-metrics dashboard covering 5 channels (funnel, failures, p50/p95 delivery latency) for on-call triage.
- Upgraded shared platform libraries to Spring Boot 4 and Java 21, and secured inbound partner webhooks with HMAC signature verification and API gateway policies.
- Developed microservices for the platform using Java and Spring Boot, including the Customer–Seller Communications Platform REST APIs.

Dec 2021
Jan 2020

Senior Software Engineer

Dell EMC · Bangalore

PROJECT **Enterprise Delivery Portal**

Java, Spring Boot, MongoDB, Node.js 10+, Angular 8, TypeScript

- Managed and led the team building a web version of a desktop (Electron-based) application.
- Developed microservices with Java and Spring Boot to deliver product requirements through product sign-off.
- Applied microservices architecture principles to design, develop, secure and test each microservice.
- Integrated security features into the microservices and their inter-service communication.
- Worked with the security team to make the application robust, including code signing and securing MQTT communication channels.
- Developed a tool to communicate with remote servers and run tools in non-GUI, Linux-based environments.

- Handled the project's deployment pipeline.

Jan 2020
Dec 2018

Software Engineer

Dell EMC (via JK Technosoft) · Bangalore

PROJECT **Backup/Recovery Product Troubleshooting Tool**

Angular 7, Node.js 8+, Express, TypeScript, Electron, HTML5, CSS/SCSS

- Developed an Electron-based desktop application using Angular and Node.js.
- Built a responsive, cross-platform, single-instance application for Windows and various Linux distributions.
- Built a secure application that uses multiple background threads to make synchronous calls.
- Developed asynchronous features using the observable pattern in Node.js.
- Developed network test and validation modules using DNS lookups, NTP servers and similar checks.
- Built rich UI components with PrimeNG and Clarity.
- Secured the application against cross-site scripting (XSS).
- Validated code quality and removed vulnerabilities using SonarQube, Fortify and Black Duck.
- Built features to make remote calls and run scripts in a secured environment.
- Loaded and validated large volumes of JSON data in batches.

Oct 2018
Oct 2017

Software Developer & Technical Trainer

NIIT Ltd. · Kolkata

PROJECT **Skill Map Engine**

Angular 5, Node.js 8+, Microservices, Spring Boot 2, Hibernate 5, Gradle 4+, JDK 1.8

- Developed client-server applications using Spring, Hibernate, REST, Maven and jQuery.
- Built secure, responsive single-page web applications using Angular and Node.js.
- Developed secure web applications using JSON, REST web service APIs and Spring Security.
- Built an application that consumes JSON from RESTful web services using JavaScript and Angular 5.
- Developed with web standards and technologies including HTML5, Bootstrap, DOM, CSS, SQL and TypeScript.
- Explored emerging JavaScript technologies such as Node.js for future product and feature development.
- Worked in a test-driven development (TDD) and CI/CD environment.
- Trained developers in server-side technologies (Node.js, J2EE) and in architectures, frameworks and design patterns for designing and refactoring web applications.

Sep 2017 – Aug 2012

Associate Software Developer

NSCT · Kolkata

PROJECT
Profile & Resume Builder
Angular 5, Node.js, Microservices, Spring Boot 2, Hibernate 4

PROJECT
E-commerce Portal (DevOps)
Spring 5, Spring Security 5, Microservices, Hibernate ORM 4

PROJECT
Collaboration Platform (Blog & Forum)
AngularJS, RESTful Web Services, Hibernate 4

- Developed Java-based web applications.
- Built rich, responsive web user interfaces using JavaScript, Bootstrap and web frameworks.
- Developed front-end web pages using HTML5, CSS3, JavaScript and AngularJS.
- Created security roles for strict control of UI elements and functions.
- Collaborated with the team on new features and UX, and built working prototypes from UI designs for cross-platform devices.
- Worked with RDBMS technology, data modelling, SQL and query optimization.
- Wrote shell scripts on Unix-based operating systems.
- Practiced test-driven development in an Agile team.

EDUCATION

MCA, Master of Computer Applications
IGNOU · 2020

PGDCA, Post Graduate Diploma in Computer Applications 74%
IGNOU · 2015

BCA, Bachelor of Computer Applications CGPA 8.97
West Bengal University of Technology · 2009–2012

Higher Secondary 78%
WBCHSE · 2009

Secondary 84.75%
WBSE · 2007